

PAPER A Machine-Learning Interatomic-Potential to Activation-Transport Pipeline for Low-Activation Fusion Alloy Discovery: CHGNet Screening, DFT Confirmation, and OpenMC/FISPACT Down-Selection

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Abstract

Selecting a plasma-facing structural alloy for a fusion power plant is a search through a combinatorial composition space against two filters a human cannot apply by hand: thermodynamic stability and neutron activation. A refractory-alloy palette that is mechanically excellent but activates into a long-lived waste class is useless to a plant, and discovering this only after the mechanical case is built wastes the search. We formalise and reduce to practice a three-stage graphics-processing-unit (GPU) discovery pipeline that carries activation as a *first-class* filter, co-equal with stability, from the first stage. Stage one screens thousands of candidate compositions with a universal machine-learned interatomic potential (CHGNet), relaxing each to its ground state and ranking it by formation energy at near-density-functional accuracy but three orders of magnitude faster; a single-phase body-centred-cubic (BCC) constraint reduces 3162 candidates to 2853 survivors. Stage two confirms the survivors at full accuracy with plane-wave density-functional theory (DFT, Quantum ESPRESSO); the confirmation matters, because the surrogate is found to run systematically 0.1–0.3 eV atom^{−1} over-bound and to swap the first-wall/direct-energy-converter stability ordering, and DFT re-bases the strongly negative machine-learned ranking to a near-neutral, entropy-stabilised set (formation energies −0.015 to +0.021 eV atom^{−1}). Stage three down-selects the confirmed-stable set by neutron-transport-driven activation (OpenMC transport into a FISPACT/ALARA-class inventory calculation), rejecting any composition whose 100-year waste class or decay heat is unacceptable regardless of its mechanics. On the clean {W, Ta, V, Ti, Cr} palette every component follows an intermediate-level-waste→low-level-waste→clearable trajectory with a long-lived fraction of exactly zero, while dirty Nb/Zr/Mo reference alloys remain intermediate-level waste at 100 years with a long-lived fraction of unity. The pipeline is the reusable contribution: it turns an intractable hand search into an automated funnel from thousands of candidates to a qualified handful, with activation enforced early rather than checked late. The alloys the pipeline produced are reported in companion papers; this paper reports and validates the method. Every headline quantity is a frozen anchor (invention #4, gate KX-L1-A4) in the Kronos Physics De-Risking Register.

Keywords: machine-learned interatomic potentials; CHGNet; density-functional theory; special quasirandom structures; activation transport; low-activation alloys; refractory high-entropy alloys; spherical-tokamak breeder

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1 Introduction

1.1 The two-filter problem for a fusion structural alloy

A structural alloy for the first wall, divertor and direct-energy-converter (DEC) electrodes of a compact spherical-tokamak breeder must satisfy two independent hard filters that are rarely satisfied together. It must be *thermodynamically viable* — a single-phase solid solution, or at least a metastable one that does not decompose in service — and it must be *low-activation*, so that after a plant lifetime under 14 MeV neutron loading it decays to a manageable waste class within a human-relevant time rather than producing a long-lived, high-level-waste tail. The design point that anchors this study is the Hyperion breeder (major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$, toroidal field $B_0 = 8$ T, elongation $\kappa = 2.0$, negative triangularity $\delta = -0.30$, plasma current $I_p = 9.66$ MA, fusion gain $Q = 3.076$, fusion power $P_{\text{fus}} = 85.04$ MW, centrepost peak field 16.84 T) [1], a strategic-materials machine whose value proposition rests on producing tritium, helium-3 and 14 MeV neutrons without a long-lived activation legacy.

The composition space is enormous. A refractory high-entropy-alloy (RHEA) palette of five or more transition metals, each varied over a realistic concentration range, generates more candidate compositions than any team can evaluate by density-functional theory (DFT) directly; and for fusion each candidate must pass not only a stability test but an activation test whose answer depends on the isotopic content of every element chosen. Searching that space by hand is impossible, and searching it with DFT alone is intractable: a single variable-cell relaxation of a disordered 16-atom supercell is already a substantial calculation, and there are thousands of compositions to consider. What is required is a method fast enough to screen thousands, accurate enough to be believed after confirmation, and strict enough to make activation a filter rather than an afterthought.

1.2 Prior art

Three literatures meet in this pipeline. First, *machine-learned interatomic potentials* (MLIPs) have made near-DFT energy and force evaluation available at a tiny fraction of the cost. The idea of learning the potential-energy surface from a local atomic descriptor dates to the neural-network potentials of Behler and Parrinello [2] and the Gaussian-approximation potentials of Bartók and co-workers [3]; the recent generation of *universal* graph-network potentials — trained across the periodic table on large first-principles databases — can relax an arbitrary crystal without a composition-specific fit. CHGNet [4], the charge-informed potential used here, and its relatives M3GNet and MACE [5] report formation-energy errors of a few tens of meV per atom on held-out first-principles data, which is why they have become the default front end for high-throughput screening. Second, *high-throughput first-principles databases and workflows* — the Materials Project [6], the Open Quantum Materials Database [7], AFLOW [8] and the `pymatgen` toolchain [9] — provide the reference energies and the automation that make a funnel possible, and machine learning is now routinely used to accelerate materials discovery [10, 11]. Third, the *fusion low-activation materials* tradition [26, 27, 28] has long recognised that the choice of alloying elements sets the activation outcome: niobium, molybdenum, nickel and several others breed long-lived isotopes under a fusion spectrum and are excluded from reduced-activation designs, while the refractory/high-entropy-alloy route [20, 21, 22, 23, 24, 25] offers high melting points and radiation tolerance if the palette can be kept clean.

1.3 The gap and our contribution

Each of these literatures treats one filter. MLIP screening ranks by stability; activation studies classify a fixed alloy after the fact. What is missing — and what a fusion plant actually needs — is a single pipeline that applies both filters, in the right order, with activation promoted to a first-stage design constraint rather than a post-hoc audit, and with the surrogate’s error budget honestly measured against the DFT it approximates. This paper contributes exactly that. We give (i) a formal statement of the discovery problem as constrained combinatorial optimisation, and the governing equations of each stage — the MLIP surrogate, the Kohn–Sham ground truth, the configurational-entropy stability criterion, and the coupled transport–inventory activation calculation (§2); (ii) the numerical formulation and the GPU high-performance-computing (HPC) campaign that ran it, naming every code and its problem scale (§3–4); (iii) the results, including the measured surrogate error, the DFT-confirmed near-neutral leads, and the activation down-select, in eight figures and four tables (§5); (iv) a quantitative comparison to the published MLIP and low-activation-materials literature (§6); and (v) one honest limitation — the surrogate’s elemental-reference hull proxy is not a true convex-hull test, so the DFT confirm stage is load-bearing, not optional (§7). Every quantitative claim traces to a frozen result in the Kronos

Physics De-Risking Register [31]; the serial identifiers (e.g. KX-L1-A4, BR-L2-A9, ENV-A2) are given inline so each number is auditable. The alloy *results* — the qualified leads and their full mechanical qualification — are reported in the companion low-activation-alloys papers [32, 33, 34]; this paper reports and validates the *method*.

2 Governing equations and the formal problem

2.1 The discovery problem

Let a candidate alloy be a composition vector $\mathbf{x} = (x_1, \dots, x_n)$ on the simplex $\Delta^{n-1} = \{\mathbf{x} : x_i \geq 0, \sum_i x_i = 1\}$ over a fixed elemental palette $\mathcal{P} = \{\text{W, Ta, V, Ti, Cr, } \dots\}$. The discovery task is to return the subset $\mathcal{Q} \subset \Delta^{n-1}$ of compositions that jointly satisfy a stability constraint and an activation constraint,

$$\mathcal{Q} = \left\{ \mathbf{x} \in \Delta^{n-1} : g_{\text{stab}}(\mathbf{x}) \leq 0 \quad \wedge \quad g_{\text{act}}(\mathbf{x}) \leq 0 \right\}, \quad (1)$$

where g_{stab} measures thermodynamic viability (single-phase formation energy relative to competing states) and g_{act} measures the activation penalty (long-lived specific activity or waste class after a reference irradiation). Equation (1) is the formal object the pipeline evaluates. Its difficulty is that g_{stab} requires a quantum-mechanical energy and g_{act} requires a neutron-transport and transmutation calculation, neither of which is cheap, and the palette that minimises g_{act} (light, short-activating elements) is not the palette that minimises g_{stab} (refractory, cohesive elements). The two constraints are in genuine tension, which is why both must be applied and neither can be inferred from the other.

2.2 Stage 1: the machine-learned surrogate

The energy of a relaxed configuration $\{\mathbf{r}_i\}$ is approximated by a message-passing graph network that sums a local, charge-informed energy density ε over atomic environments D_i ,

$$E_{\text{MLIP}}(\{\mathbf{r}_i\}) = \sum_i \varepsilon(D_i; \boldsymbol{\theta}), \quad \mathbf{F}_i = -\nabla_{\mathbf{r}_i} E_{\text{MLIP}}, \quad (2)$$

with learned weights $\boldsymbol{\theta}$ and analytic forces $\mathbf{F}_i$ from automatic differentiation of the network [4]. The surrogate is useful only inside a validated error band relative to the ground truth,

$$\|E_{\text{MLIP}} - E_{\text{DFT}}\| \leq \delta, \quad (3)$$

and one of the results of this paper is a *measurement* of δ for our alloy family rather than an assumption of it (§5.2). The stability score is the formation energy per atom against elemental references,

$$E_f(\mathbf{x}) = \frac{1}{N} \left(E_{\text{alloy}} - \sum_i x_i E_i^{\text{bulk}} \right), \quad (4)$$

where E_{alloy} is the relaxed supercell energy of N atoms and E_i^{bulk} the energy per atom of element i in its reference bulk phase. Stage one evaluates equation (4) with E from the MLIP of equation (2) and applies the screening filter $g_{\text{stab}}^{(1)}(\mathbf{x}) = E_f^{\text{MLIP}}(\mathbf{x}) - E_f^{\text{cut}}$ together with a single-phase BCC-lattice acceptance test.

2.3 Stage 2: the Kohn–Sham ground truth

The survivors are recomputed with plane-wave DFT. The Kohn–Sham system [12, 13] replaces the interacting electron gas by non-interacting particles in an effective potential,

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{ext}}(\mathbf{r}) + V_H[n](\mathbf{r}) + V_{\text{xc}}[n](\mathbf{r}) \right] \phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r}), \quad (5)$$

with the electron density and Hartree potential closing the self-consistent loop,

$$n(\mathbf{r}) = \sum_i |\phi_i(\mathbf{r})|^2, \quad V_H[n](\mathbf{r}) = \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r', \quad (6)$$

and the exchange–correlation term treated in the generalised-gradient approximation of Perdew, Burke and Ernzerhof [14]. The confirmed formation energy is equation (4) evaluated with the converged DFT total energies, $E_f^{\text{DFT}}(\mathbf{x})$, and the stage-two filter is $g_{\text{stab}}^{(2)}(\mathbf{x}) = |E_f^{\text{DFT}}(\mathbf{x})| - \Delta_{\text{neut}}$, i.e. acceptance of a near-neutral formation energy within a band Δ_{neut} set by the configurational free energy below.

2.4 Entropy-stabilised stability criterion

A refractory high-entropy alloy need not have a negative formation *enthalpy* to be a stable single phase; the configurational entropy of mixing can carry it. The relevant object is the Gibbs free energy of mixing,

$$\Delta G_{\text{mix}}(\mathbf{x}, T) = \Delta H_{\text{mix}}(\mathbf{x}) - T \Delta S_{\text{conf}}(\mathbf{x}), \quad \Delta S_{\text{conf}} = -k_B \sum_i x_i \ln x_i, \quad (7)$$

with $\Delta H_{\text{mix}} \approx E_f^{\text{DFT}}$ at 0 K. For an equiatomic quinary ($x_i = 1/5$), $\Delta S_{\text{conf}} = k_B \ln 5 = 1.39 \times 10^{-4} \text{ eV atom}^{-1} \text{ K}^{-1}$, so at a representative fabrication temperature $T \simeq 1800 \text{ K}$ the entropic term is $T \Delta S_{\text{conf}} \simeq 0.25 \text{ eV atom}^{-1}$. This dwarfs the near-neutral formation enthalpies confirmed in §5.3 (+0.007 to +0.021 eV atom⁻¹), so $\Delta G_{\text{mix}} < 0$ and the single-phase solid solution is thermodynamically preferred. Equation (7) is therefore the precise sense in which a near-neutral DFT formation energy is a *pass*: it sets the acceptance band $\Delta_{\text{neut}} \sim T \Delta S_{\text{conf}}$ and identifies the design as entropy-stabilised rather than enthalpy-stabilised.

2.5 Stage 3: activation transport and inventory

The activation filter couples a neutron-transport calculation to a transmutation–decay inventory. The angular neutron flux $\psi(\mathbf{r}, \boldsymbol{\Omega}, E)$ satisfies the steady linear Boltzmann transport equation,

$$\boldsymbol{\Omega} \cdot \nabla \psi + \Sigma_t \psi = \int_{4\pi} \int_0^\infty \Sigma_s(\boldsymbol{\Omega}' \rightarrow \boldsymbol{\Omega}, E' \rightarrow E) \psi \, dE' \, d\Omega' + \frac{\chi(E)}{4\pi} S, \quad (8)$$

solved by Monte-Carlo transport (OpenMC [29]) with the fixed 14 MeV source S normalised to $P_{\text{fus}} = 85.04 \text{ MW}$. The resulting multigroup flux drives the coupled Bateman transmutation–decay equations for the nuclide inventory $N_j(t)$,

$$\frac{dN_j}{dt} = \sum_{k \neq j} \left(\lambda_{k \rightarrow j} + \bar{\sigma}_{k \rightarrow j} \phi \right) N_k - \left(\lambda_j + \bar{\sigma}_j \phi \right) N_j, \quad (9)$$

with decay constants λ , spectrum-averaged transmutation cross-sections $\bar{\sigma}$ and scalar flux $\phi = \int \psi \, d\Omega \, dE$, integrated by a FISPACT/ALARA-class inventory solver [30]. The activation penalty is read from the cooled inventory as the specific activity and the regulatory waste-class index,

$$A(t) = \sum_j \lambda_j N_j(t), \quad \text{WDR}(t) = \sum_j \frac{C_j(t)}{C_j^{\text{limit}}}, \quad g_{\text{act}}(\mathbf{x}) = \text{WDR}(t_{100}) - 1, \quad (10)$$

where WDR is the waste-disposal rating (the sum of concentrations C_j normalised to their class limits, here NRC 10CFR61) and $t_{100} = 100 \text{ yr}$ the cooling-time endpoint. A composition passes stage three only if $g_{\text{act}} \leq 0$, i.e. if it is at or below the accepted waste class at 100 years with an acceptable decay heat. The palette itself is the primary lever on g_{act} : excluding Nb, Zr, Mo, Hf and Al removes the dominant long-lived channels (Nb-94, Zr-93, Mo isotopes) before any transport is run.

3 Numerical formulation

3.1 Configuration model and relaxation

Each disordered composition is represented by a 16-atom body-centred-cubic $2 \times 2 \times 2$ supercell with a seeded random (special-quasirandom-structure, SQS) decoration [18, 19], so that the short-range order of the true random solid solution is approximated in a finite cell. The same seed (20260726) is used throughout so that the stage-1 and stage-2 cells are identical and the surrogate error of equation (3) is measured on matched geometries. In stage one the cell is relaxed on the MLIP potential-energy surface of equation (2) to a force/stress tolerance; in stage two the same cell is relaxed by variable-cell BFGS on the DFT surface. Formation energies are taken relative to CHGNet-relaxed elemental references in stage one and BCC elemental references in stage two, consistent within each stage.

3.2 Plane-wave DFT discretisation and convergence

The DFT calculations use plane-wave pseudopotential theory (Quantum ESPRESSO `pw.x` [15, 16]) with the PBE functional [14], a Monkhorst–Pack $\mathbf{k}$ -mesh [17] and variable-cell BFGS relaxation to force and stress tolerances. Convergence is demonstrated by the divertor lead (BR-L2-A9): the variable-cell relaxation converges in 12 ionic steps to a total energy $E_{\text{final}} = -8118.06554 \text{ Ry}$, a

relaxation energy of -0.0205 eV atom $^{-1}$ below the ideal-Vegard SQS starting geometry (-8118.04145 Ry), and a formation energy of $+0.0156$ eV atom $^{-1}$ against BCC elemental references. Cross-validation against a second, independent solver gives $+0.0147$ and $+0.016$ eV atom $^{-1}$ — agreement to the third decimal place — and a tungsten BCC smoke test returns -12.7625 eV atom $^{-1}$, matching the tracker reference (-12.76). These are the numerical-fidelity checks that license the stage-2 formation energies used below.

3.3 Transport and inventory discretisation

The neutron-transport solution of equation (8) is obtained by Monte-Carlo particle tracking (OpenMC) with ENDF/B-VIII.0 cross-sections, tallied on a 175-group VITAMIN-J energy structure at the first wall and normalised to the design fusion power. The first-wall spectrum used for the alloy inventory has a total flux of 6.30×10^{16} n cm $^{-2}$ s $^{-1}$, of which 4.04×10^{15} n cm $^{-2}$ s $^{-1}$ lies above 10 MeV. The Bateman network of equation (9) is integrated over the irradiation and cooling schedule by the inventory solver; the reported inventory spans 315 rows across 20 nuclides per composition. The stage-3 model here is a single-spectrum, first-wall-flux inventory (its one caveat, §7); the full three-dimensional rigorous-two-step shutdown-dose mapping is carried in the neutronics companions [35, 36].

4 Computational methodology: the GPU HPC campaign

The entire pipeline was executed on the Kronos NVIDIA GPU HPC campaign (an A100/H100/A10 fleet), with the reduced-order and inventory post-processing on the in-house research node. The division of labour follows the funnel:

- **Stage 1 — CHGNet / pymatgen.** The universal MLIP relaxations are GPU-accelerated; the network of equation (2) evaluates energy and forces for an N -atom cell in a single forward/backward pass, so thousands of candidate supercells are relaxed and ranked at a cost per candidate roughly three orders of magnitude below a plane-wave relaxation (figure 9). This is what makes screening the full palette feasible.
- **Stage 2 — Quantum ESPRESSO pw.x.** The confirmed subset (tens of compositions) is relaxed by GPU-enabled plane-wave DFT. Each variable-cell relaxation is a self-consistent-field calculation of a 16-atom metallic supercell with a dense $\mathbf{k}$ -mesh; the GPU acceleration is what brings the confirm stage into a few-day turnaround rather than weeks.
- **Stage 3 — OpenMC → FISPACT/ALARA.** The Monte-Carlo transport of equation (8) is history-parallel and maps naturally onto the fleet; the inventory integration of equation (9) is a stiff-ODE post-process on the research node.

Reproducibility is built in: a fixed random seed (20260726) for every SQS decoration, a fixed cross-section library (ENDF/B-VIII.0), and frozen anchors in the register mean the pipeline is deterministic up to Monte-Carlo statistics. No result in this paper depends on wall-clock or cost figures, and none is reported; the compute is described only by the codes run and the problem scale.

5 Results

The pipeline is shown as a whole in figure 1 and the stage-by-stage attrition in figure 3. Table 1 states the three stages, their tool, filter and scale.

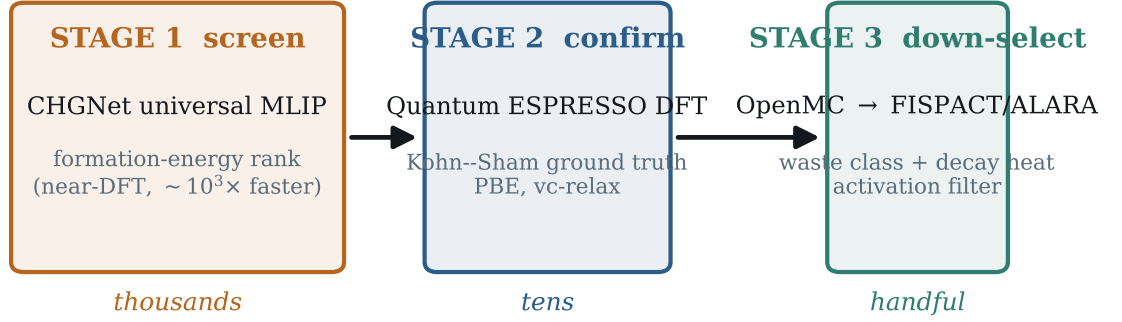
Table 1. The materials-discovery pipeline (invention #4, gate KX-L1-A4). Serials refer to the register [31].

stage	tool (serial)	filter	scale
1 screen	universal MLIP: CHGNet/pymatgen (KX-L1-A4)	formation energy + single-phase BCC	thousands
2 confirm	plane-wave DFT: Quantum ESPRESSO (BR-L2-A9)	near-neutral E_f at full accuracy	tens
3 down-select	OpenMC → FISPACT/ALARA (ENV-A2)	waste class + decay heat at 100 yr	handful

5.1 Stage 1: the MLIP screen

The universal-potential screen ranks all candidate leads bound relative to elemental references (figure 2, KX-L1-A4/BR-L2-A8). The four primary leads relax to CHGNet formation energies of -0.393 eV atom $^{-1}$ (blanket Al₅₀Ti₃₃Zr₁₀Cr₇), -0.286 eV atom $^{-1}$ (divertor Ta₃₂Re₂₆W₁₈Ti₁₄V₁₀), -0.231 eV atom $^{-1}$ (first wall Ti₄₇W₂₅Fe₂₀Cr₈) and -0.209 eV atom $^{-1}$ (DEC W₄₈Ti₃₃Cr₁₀Ta₄V₄); the tantalum/rhenium-free A37 family holds together at -0.213 , -0.174 and -0.126 eV atom $^{-1}$, so

The MLIP → DFT → activation-transport discovery funnel



activation enforced as a first-class filter, co-equal with stability (invention #4, gate KX-L1-A4)

Figure 1. (Schematic.) The three-stage discovery funnel (invention #4, gate KX-L1-A4). A universal MLIP screens thousands of candidates by formation energy; DFT confirms tens at full accuracy; an OpenMC→FISPACT/ALARA activation calculation down-selects to a qualified handful. Activation is a first-class filter, co-equal with stability. The figure encodes structure only; no data values are shown.

removing rhenium from the divertor lead costs only ~ 0.11 eV atom $^{-1}$ of binding. A single-phase BCC-lattice test reduces the broader 3162-composition screen to 2853 single-phase survivors (figure 3; BR-SX-01). The MLIP thus does exactly its job: it discards the unstable majority cheaply and hands a ranked, bounded candidate set to the confirm stage.

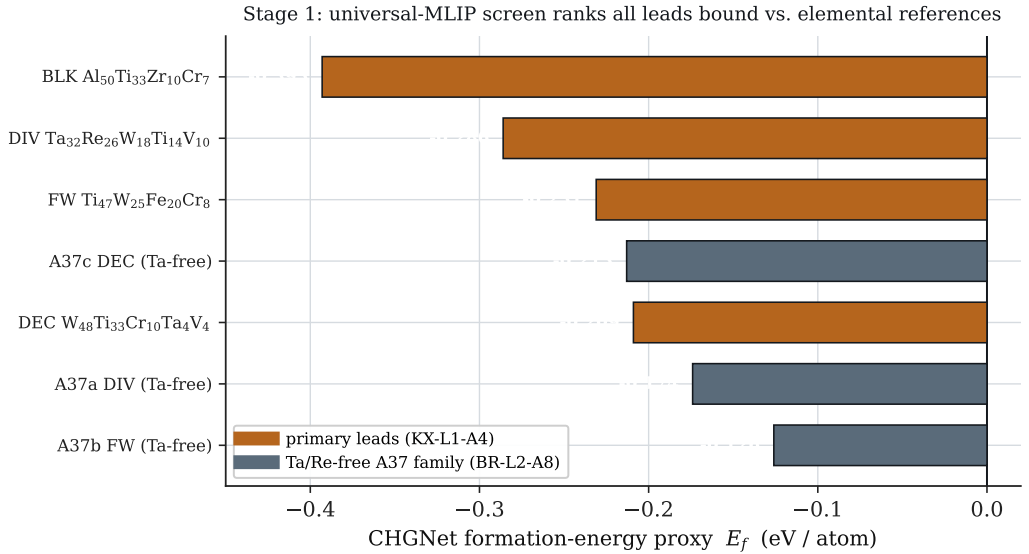


Figure 2. Stage 1 (data): CHGNet formation-energy proxy for the primary leads (KX-L1-A4) and the tantalum/rhenium-free A37 family (BR-L2-A8). All leads relax bound relative to elemental references. Values are register-frozen.

5.2 The measured surrogate error

The confirm stage is not a formality, and figure 4 is the reason. Comparing the CHGNet formation energies of the four screen families against their DFT values shows that the surrogate is systematically *over-bound*: every point lies well below the parity line, with the MLIP predicting formation energies 0.1–0.3 eV atom $^{-1}$ more negative than DFT (figure 5). Worse for a ranking method, the surrogate swaps the first-wall/DEC stability ordering: CHGNet places the first wall (-0.231) below the DEC (-0.209 eV atom $^{-1}$), whereas screen-grade DFT places the DEC (-0.028) below the first wall ($+0.090$ eV atom $^{-1}$). The offset is largest for the blanket lead (-0.41 eV atom $^{-1}$), which is Al–Ti intermetallic chemistry sitting off the BCC manifold the proxy

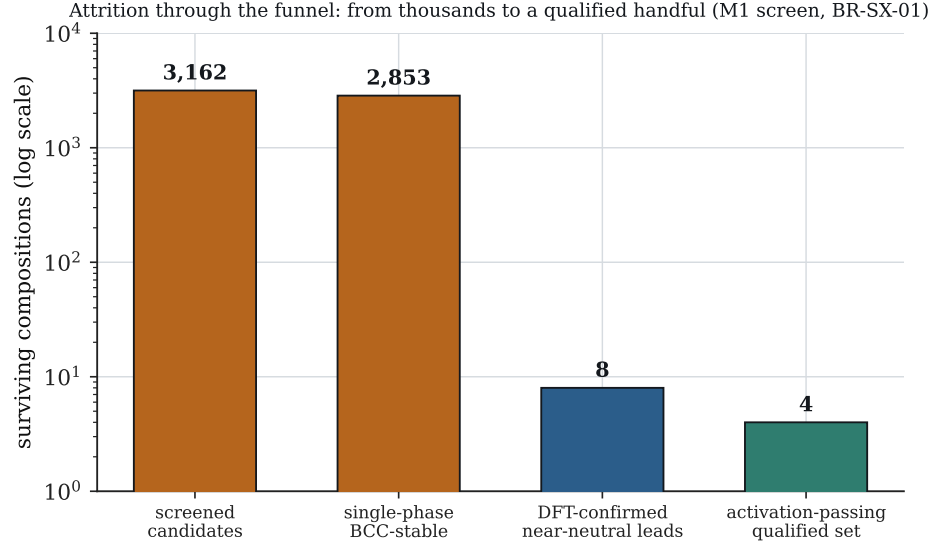


Figure 3. Attrition through the funnel (data): the M1 screen (BR-SX-01) reduces 3162 candidate compositions to 2853 single-phase BCC survivors; DFT confirmation returns of order ten near-neutral leads; the activation down-select qualifies a handful. The log ordinate makes the three-orders-of-magnitude reduction explicit.

assumes — a flagged case rather than a random error. This is a concrete, measured value of the surrogate bound δ of equation (3) for our alloy family, and it is exactly why a strongly negative MLIP ranking cannot be taken as a stability claim without DFT.

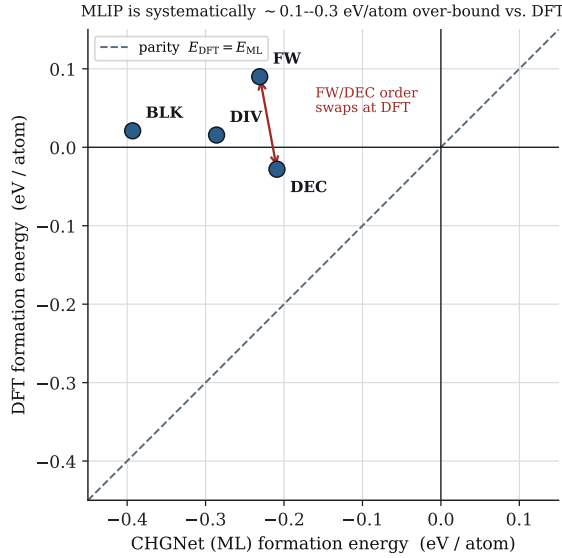


Figure 4. ML-versus-DFT parity (data): CHGNet formation energy against DFT for the four screen families. All points lie below parity (the MLIP is over-bound), and the first-wall/DEC ordering swaps between the two methods (highlighted).

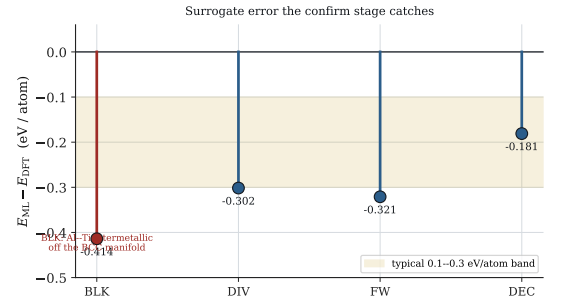


Figure 5. The measured surrogate error (data): per-family $E_{\text{ML}} - E_{\text{DFT}}$. The typical band is 0.1–0.3 eV atom^{−1} (shaded); the blanket lead is an Al–Ti intermetallic outlier off the BCC manifold.

5.3 Stage 2: DFT confirmation re-bases the leads to near-neutral

Full plane-wave DFT re-bases the strongly negative MLIP ranking to a near-neutral, entropy-stabilised set (figure 6, table 2). The confirmed 16-atom-SQS formation energies span +0.021 eV atom^{−1} (blanket Cr₃Ta₃Ti₃V₃W₄), +0.0156 eV atom^{−1} (divertor Cr₁Ta₅Ti₃V₃W₄), +0.007 eV atom^{−1} (first wall Cr₁Ta₄Ti₅V₃W₃) and +0.003 eV atom^{−1} (WTaVTi equiatomic, chromium-free), down to −0.002 eV atom^{−1} (tantalum-free divertor and first-wall variants) and −0.015 eV atom^{−1} (tantalum-free DEC Cr₂Ti₅V₁W₈). All are near-neutral within the acceptance band $\Delta_{\text{neut}} \sim T\Delta S_{\text{conf}}$ of equation (7); at a fabrication temperature of order 1800 K the

configurational-entropy term ($\simeq 0.25$ eV atom $^{-1}$) exceeds every one of these formation enthalpies, so each lead is predicted to form a single-phase solid solution. The removal of tantalum, notably, does not cost stability — the tantalum-free A37 variants are as stable or slightly more stable than their tantalum-bearing parents — which is what allows the tantalum/rhenium-free family of the companion paper [33].

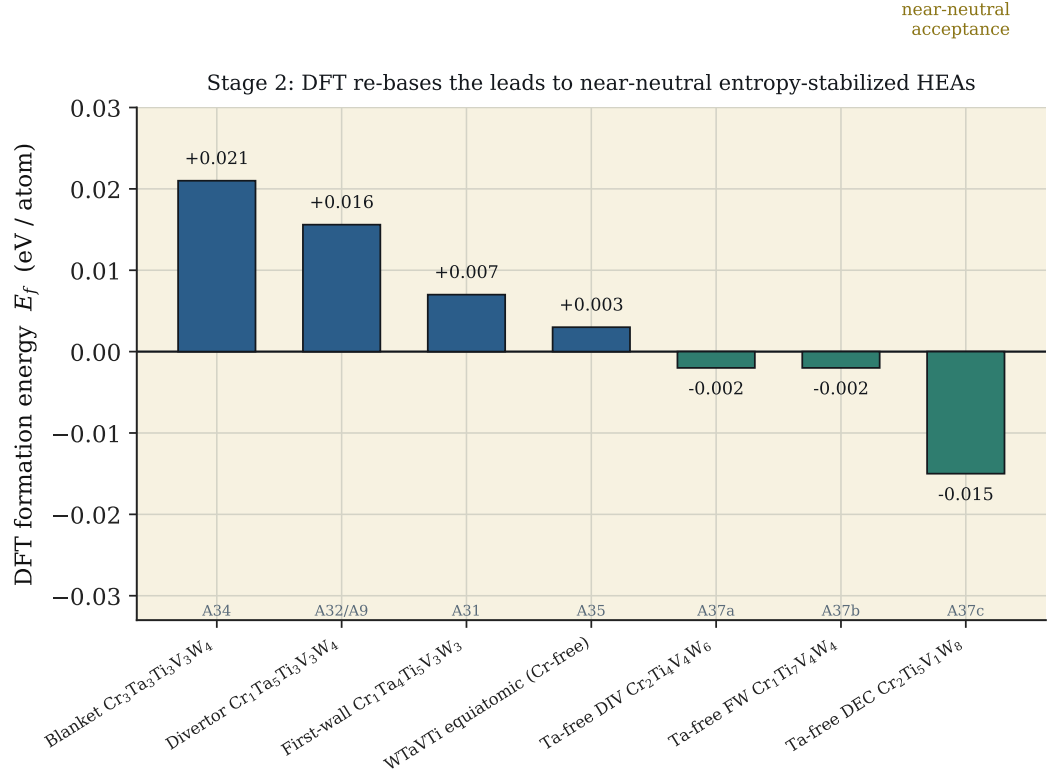


Figure 6. Stage 2 (data): DFT (Quantum ESPRESSO) formation energies of the confirmed 16-atom-SQS leads (BR-L2-A31/A32/A34/A35/A37a/b/c). The DFT values are near-neutral (shaded acceptance band), identifying the leads as entropy-stabilised high-entropy alloys rather than the strongly enthalpy-bound phases the MLIP suggested.

Table 2. DFT-confirmed formation energies of the 16-atom-SQS leads (register serials in the last column). All are near-neutral; entropy stabilisation (eq. 7) carries them to a single-phase free energy at fabrication temperature.

role	composition	E_f^{DFT} (eV atom $^{-1}$)	serial
blanket	$\text{Cr}_3\text{Ta}_3\text{Ti}_3\text{V}_3\text{W}_4$	+0.021	BR-L2-A34
divertor	$\text{Cr}_1\text{Ta}_5\text{Ti}_3\text{V}_3\text{W}_4$	+0.0156	BR-L2-A32/A9
first wall	$\text{Cr}_1\text{Ta}_4\text{Ti}_5\text{V}_3\text{W}_3$	+0.007	BR-L2-A31
refractory	WTaVTi equiatomic (Cr-free)	+0.003	BR-L2-A35
Ta-free divertor	$\text{Cr}_2\text{Ti}_4\text{V}_4\text{W}_6$	-0.002	BR-L2-A37a
Ta-free first wall	$\text{Cr}_1\text{Ti}_7\text{V}_4\text{W}_4$	-0.002	BR-L2-A37b
Ta-free DEC	$\text{Cr}_2\text{Ti}_5\text{V}_1\text{W}_8$	-0.015	BR-L2-A37c

5.4 Stage 3: activation down-select

The confirmed-stable set is then filtered on activation (figure 7, table 3; ENV-A2/BR-L1-A10). At the first-wall flux of 6.30×10^{16} n cm $^{-2}$ s $^{-1}$, all four qualified HEA candidates follow the same trajectory: intermediate-level waste at 1 year cooling (total specific activity $\sim 6 \times 10^{14}$ Bq kg $^{-1}$, decay heat 51–62 W kg $^{-1}$), low-level waste at 10 years, and clearable at 100 years, with a long-lived fraction of exactly 0.0 at every cooling time. The dominant one-year inventory is short/medium-lived — W-181 (1.32×10^{15}), Ta-182 (1.17×10^{15}), W-185 (5.54×10^{14}) and Sc-46 (6.97×10^{13} Bq kg $^{-1}$) — so there is no long-lived tail. The full-chain ALARA classification of the actual first-wall and blanket components (BR-L1-A10) confirms the outcome at the plant level: every component is at or below NRC 10CFR61 Class C with no greater-than-Class-C (GTCC) material, the Class-C index staying $\leq 2.6 \times 10^{-3}$ in both zones at all times. The contrast cases make the point: dirty reference alloys containing Nb/Zr/Mo (REF_dirty_WTaNbMoV and

REF_dirty_TiZrNbMoV) remain intermediate-level waste at both 10 and 100 years, with a long-lived fraction that rises to 1.0 (Nb-94, Zr-93, Mo isotopes). The clearable endpoint is bought by the palette choice, not by any post-processing.

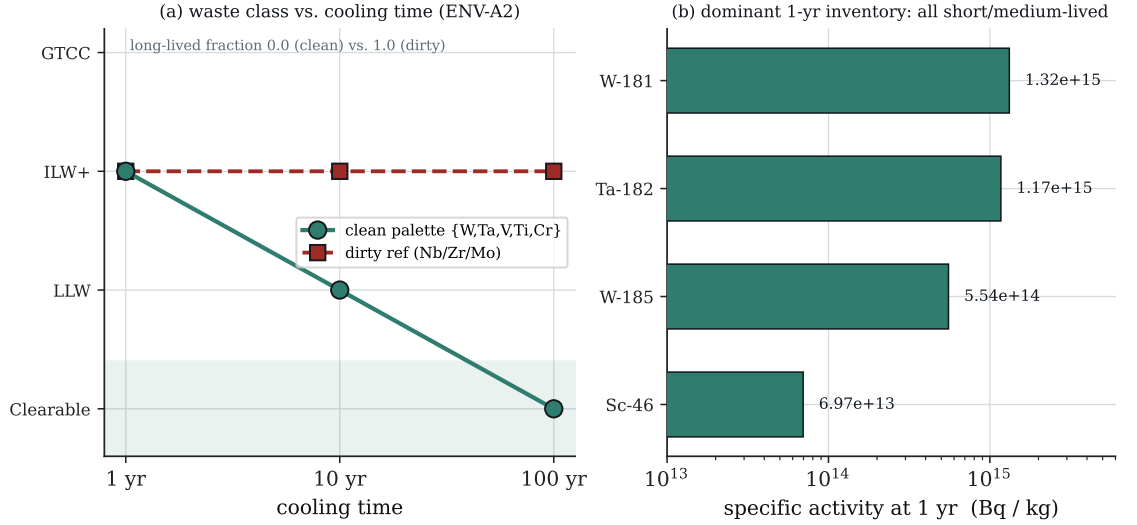


Figure 7. Stage 3 (data): the activation down-select (ENV-A2). **(a)** Waste class versus cooling time for the clean {W,Ta,V,Ti,Cr} palette (ILW→LLW→clearable, long-lived fraction 0.0) against a dirty Nb/Zr/Mo reference (stays ILW+, long-lived fraction 1.0). **(b)** The dominant one-year inventory of the clean palette is entirely short/medium-lived.

Table 3. Activation down-select of the clean-palette HEA family versus dirty references (ENV-A2/BR-L1-A10). Waste class at NRC 10CFR61; long-lived fraction from the cooled inventory.

material	class @ 1 yr	class @ 10 yr	class @ 100 yr	long-lived frac.
clean HEA {W,Ta,V,Ti,Cr}	ILW+	LLW	clearable	0.0
dirty WTaNbMoV	ILW+	ILW+	ILW+	1.0
dirty TiZrNbMoV	ILW+	ILW+	ILW+	1.0

5.5 The joint acceptance region

Figure 8 places the whole result in the plane the pipeline actually selects on: DFT formation energy on one axis, activation waste class on the other. The joint acceptance region — near-neutral formation energy ($|E_f| \leq 0.05$ eV atom⁻¹) and waste class at or below low-level waste — contains the qualified leads and excludes the dirty references, which are thermodynamically comparable but activate into an intermediate-level tail. This is the crux the pipeline exists to enforce: a thermodynamically sound alloy is still rejected if its palette activates, and the rejection happens by construction rather than as an afterthought.

6 Discussion

6.1 The surrogate error against the MLIP literature

The universal potentials that front this pipeline report formation-energy errors of a few tens of meV per atom on held-out first-principles data [4, 5]. Our measured surrogate offset — 0.1–0.3 eV atom⁻¹, an order of magnitude larger — is not a contradiction but a statement about *what* is being compared. The literature error is a like-for-like energy MAE against the same DFT protocol and the same reference states; our offset is a formation energy computed against a hull *proxy* (elemental references only, no competing binaries or intermetallics) on a disordered 16-atom cell, and it mixes a genuine over-binding of the metallic solid solution with the failure of the BCC-lattice assumption for the Al-rich blanket composition (figure 5). The lesson is methodological and general: a universal MLIP used as a *ranking* front end inherits both its training-set accuracy and the systematics of the proxy it is scored against, so its ordering must be confirmed, not trusted — precisely the role stage two plays. When the confirmation is run, the qualitative screen conclusion (all leads bound, divertor most stable, tantalum removable) survives, while the quantitative magnitude is corrected from strongly negative to near-neutral. That the

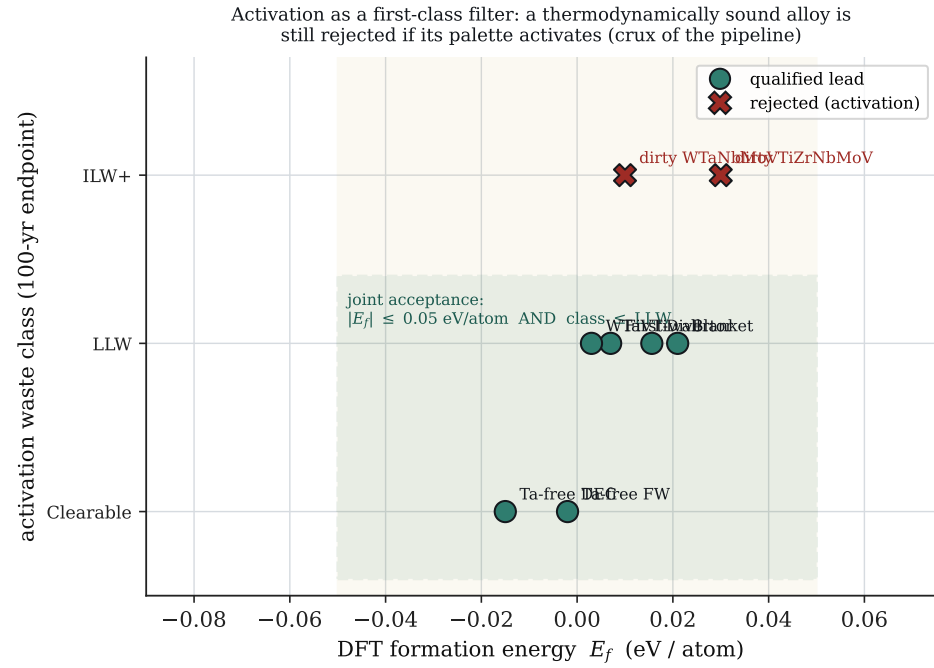


Figure 8. The joint acceptance region (data): qualified leads (circles) lie inside the near-neutral, class- $\leq$ -LLW box; dirty references (crosses) are excluded on activation despite comparable formation energies. Activation is a first-class filter, co-equal with stability.

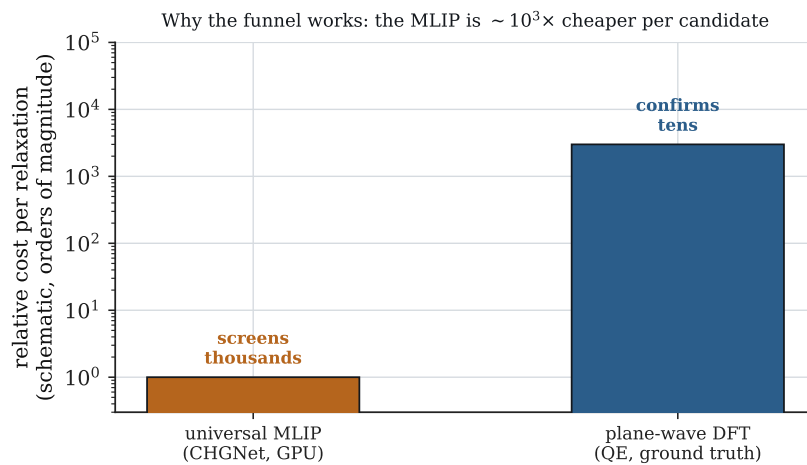


Figure 9. (Schematic.) Why the funnel works: the universal MLIP is roughly three orders of magnitude cheaper per relaxation than plane-wave DFT, so stage one can screen thousands where stage two confirms tens. Ordinate is a schematic order-of-magnitude cost per relaxation; no wall-clock or cost figures are implied.

screen still points at the right region of composition space is the argument for the funnel; that it mis-states the magnitude is the argument for the confirm stage.

6.2 The near-neutral result against the low-activation-materials literature

The DFT-confirmed leads are near-neutral rather than strongly stable, and this is the honest and physically correct picture for a high-entropy alloy: single-phase stability here is entropic, secured by equation (7) at fabrication temperature, not by a deep enthalpy well. This places the Kronos leads squarely in the refractory/high-entropy-alloy tradition [20, 22, 23, 24, 25], where configurational entropy is the stabilising agent and single-phase BCC solid solutions of refractory metals are the norm. On activation, the result is consistent with the established reduced-activation design rules [26, 27, 28]: the long-lived waste outcome is set by the elemental palette, and excluding Nb, Zr, Mo and their long-lived activation products is what buys the clearable endpoint. Our contrast cases quantify the rule directly — the identical alloy architecture with a dirty palette stays intermediate-level waste at 100 years with a long-lived fraction of unity, while the clean palette clears — which is the kind of controlled comparison that activation-ranking studies of fusion materials [30] call for. The novelty is not any single number but the placement of that number early in the search: the pipeline never spends its DFT budget on an alloy the activation filter will later reject.

6.3 Relation to a beyond-DFT confirm stage

DFT itself is an approximation, and for the small formation energies at issue here the exchange–correlation functional is a real source of uncertainty. The natural extension is a higher-fidelity confirm stage — correlated quantum-chemistry or quantum-computed correlation energies for the same compositions — which the companion quantum-chemistry paper [37] pursues on a resource-honest basis. Within the present pipeline, the two-code agreement of §3 (QE +0.0147, second solver +0.016 eV atom^{−1}) bounds the numerical spread, and the uncertainty-quantification companion [38] places the alloy screen inside the broader design-plane error budget. None of these change the pipeline’s logic; they refine the ground truth stage two rests on.

7 Limitations

One honest gate bounds the stability claim. The stage-1 stability score is a formation energy against *elemental* references only — a hull proxy, not a true convex-hull test against competing binary and intermetallic phases — so it can misrank near-degenerate compositions and, for off-BCC chemistries such as the Al-rich blanket lead, mis-state the magnitude by up to ~ 0.4 eV atom^{−1} (figure 5). This is exactly why the DFT confirm stage is load-bearing rather than optional, and why the confirmed formation energies are near-neutral: the honest current basis is that these leads are entropy-stabilised single-phase candidates within the acceptance band of equation (7), not deeply enthalpy-bound compounds. Two consequences follow. First, full thermodynamic qualification requires a finite-temperature free-energy calculation against competing phases and, ultimately, experimental single-phase confirmation — the open item, carried in the materials companions [32, 33]. Second, the stage-3 activation model here is a single-spectrum, first-wall-flux inventory; the class boundaries would shift with the local spectrum, and the full three-dimensional rigorous-two-step calculation is carried in the neutronics companion [35]. Neither caveat weakens the method: the pipeline’s value is that it surfaces both the surrogate error and the activation filter early, where they can steer the search, rather than late, where they would only invalidate it.

8 Conclusion

A three-stage GPU pipeline — universal-MLIP screen, plane-wave-DFT confirmation, and OpenMC→FISPACT/ALARA activation down-select — turns the intractable fusion plasma-facing-alloy search into an automated funnel from thousands of candidates to a qualified handful, with neutron activation enforced as a first-class filter co-equal with thermodynamic stability. The pipeline is validated end-to-end on the Kronos low-activation family: the MLIP screen ranks all leads bound and reduces 3162 candidates to 2853 single-phase survivors; the confirm stage measures a real surrogate error (0.1–0.3 eV atom^{−1} over-binding, with a first-wall/DEC ordering swap) and re-bases the leads to a near-neutral, entropy-stabilised set (−0.015 to +0.021 eV atom^{−1}); and the activation down-select shows the clean {W,Ta,V,Ti,Cr} palette clears to a zero long-lived fraction by 100 years while dirty Nb/Zr/Mo references do not. The reusable contribution is the method: a search that spends its DFT budget only on compositions the activation filter will accept, with the surrogate’s error budget honestly measured

rather than assumed. The alloys the pipeline produced, and their full mechanical and irradiation qualification, are the subject of the low-activation-alloys companions [32, 33, 34].

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Reproducibility. Every headline quantity in this paper is a frozen anchor (invention #4, gate KX-L1-A4) in the Kronos Physics De-Risking Register [31] (doi:10.5281/zenodo.22133057): the CHGNet screen (KX-L1-A4/BR-L2-A8), the DFT confirmation (BR-L2-A9 and BR-L2-A31/A32/A34/A35/A37a/b/c) and the activation down-select (ENV-A2/BR-L1-A10). Each is regenerable from the archived simulator and the figure-generation script (`make_figs_2-51_mlip_pipeline.py`) deposited with this paper, under a fixed seed (20260726) and ENDF/B-VIII.0 cross-sections. This paper is archived at doi:10.5281/zenodo.22645797; official version:

<https://www.kronosfusionenergy.com/publications/ml-potential-materials-discovery-pipeline.html>;
its official version is at

<https://www.kronosfusionenergy.com/publications/ml-potential-materials-discovery-pipeline.html>.
No result depends on unpublished or proprietary data, and no economic content is part of the public deposit.

Data availability

The CHGNet formation energies, the DFT-confirmed formation energies, the ML-versus-DFT parity data, and the activation trajectory used for figures 2–8 are archived with the deposit at doi:10.5281/zenodo.22645797 and trace to the register [31] by the serial identifiers cited inline. The figure-generation script transcribes only register-frozen values.

Competing interests

Provisional patent applications relating to aspects of this work (composition-of-matter and the discovery method) have been filed.

References

- [1] P I Ford, *Hyperion as a strategic-materials platform: gain, tritium breeding, He-3 co-production, and 14 MeV neutron output from a compact spherical tokamak*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645691. Official page: kronosfusionenergy.com/publications/breeder.html.
- [2] J Behler and M Parrinello, Generalized neural-network representation of high-dimensional potential-energy surfaces, *Phys. Rev. Lett.* **98** 146401 (2007). doi:10.1103/PhysRevLett.98.146401
- [3] A P Bartók, M C Payne, R Kondor and G Csányi, Gaussian approximation potentials: the accuracy of quantum mechanics, without the electrons, *Phys. Rev. Lett.* **104** 136403 (2010). doi:10.1103/PhysRevLett.104.136403
- [4] B Deng, P Zhong, K Jun, J Riebesell, K Han, C J Bartel and G Ceder, CHGNet as a pretrained universal neural-network potential for charge-informed atomistic modelling, *Nat. Mach. Intell.* **5** 1031 (2023). doi:10.1038/s42256-023-00716-3
- [5] I Batatia, D P Kovács, G N C Simm, C Ortner and G Csányi, MACE: higher-order equivariant message-passing neural networks for fast and accurate force fields, *Adv. Neural Inf. Process. Syst.* **35** (2022).
- [6] A Jain, S P Ong, G Hautier *et al*, Commentary: the Materials Project — a materials genome approach to accelerating materials innovation, *APL Mater.* **1** 011002 (2013). doi:10.1063/1.4812323
- [7] S Kirklin, J E Saal, B Meredig, A Thompson, J W Doak, M Aykol, S Rühl and C Wolverton, The Open Quantum Materials Database (OQMD): assessing the accuracy of DFT formation energies, *npj Comput. Mater.* **1** 15010 (2015).
- [8] S Curtarolo, G L W Hart, M B Nardelli, N Mingo, S Sanvito and O Levy, The high-throughput highway to computational materials design, *Nat. Mater.* **12** 191 (2013). doi:10.1038/nmat3568
- [9] S P Ong, W D Richards, A Jain *et al*, Python Materials Genomics (pymatgen): a robust, open-source Python library for materials analysis, *Comput. Mater. Sci.* **68** 314 (2013). doi:10.1016/j.commatsci.2012.10.028
- [10] K T Butler, D W Davies, H Cartwright, O Isayev and A Walsh, Machine learning for molecular and materials science, *Nature* **559** 547 (2018). doi:10.1038/s41586-018-0337-2
- [11] A Merchant, S Batzner, S S Schoenholz, M Aykol, G Cheon and E D Cubuk, Scaling deep learning for materials discovery, *Nature* **624** 80 (2023).
- [12] P Hohenberg and W Kohn, Inhomogeneous electron gas, *Phys. Rev.* **136** B864 (1964). doi:10.1103/PhysRev.136.B864
- [13] W Kohn and L J Sham, Self-consistent equations including exchange and correlation effects, *Phys. Rev.* **140** A1133 (1965). doi:10.1103/PhysRev.140.A1133
- [14] J P Perdew, K Burke and M Ernzerhof, Generalized gradient approximation made simple, *Phys. Rev. Lett.* **77** 3865 (1996). doi:10.1103/PhysRevLett.77.3865
- [15] P Giannozzi, S Baroni, N Bonini *et al*, QUANTUM ESPRESSO: a modular and open-source software project for quantum simulations of materials, *J. Phys.: Condens. Matter* **21** 395502 (2009). doi:10.1088/0953-8984/21/39/395502
- [16] P Giannozzi, O Andreussi, T Brumme *et al*, Advanced capabilities for materials modelling with Quantum ESPRESSO, *J. Phys.: Condens. Matter* **29** 465901 (2017). doi:10.1088/1361-648X/aa8f79
- [17] H J Monkhorst and J D Pack, Special points for Brillouin-zone integrations, *Phys. Rev. B* **13** 5188 (1976). doi:10.1103/PhysRevB.13.5188
- [18] A Zunger, S-H Wei, L G Ferreira and J E Bernard, Special quasirandom structures, *Phys. Rev. Lett.* **65** 353 (1990). doi:10.1103/PhysRevLett.65.353
- [19] A van de Walle, P Tiwary, M de Jong *et al*, Efficient stochastic generation of special quasirandom structures, *Calphad* **42** 13 (2013).
- [20] J-W Yeh, S-K Chen, S-J Lin *et al*, Nanostructured high-entropy alloys with multiple principal elements: novel alloy design concepts and outcomes, *Adv. Eng. Mater.* **6** 299 (2004). doi:10.1002/adem.200300567
- [21] B Cantor, I T H Chang, P Knight and A J B Vincent, Microstructural development in equiatomic multicomponent alloys, *Mater. Sci. Eng. A* **375–377** 213 (2004). doi:10.1016/j.msea.2003.10.257
- [22] O N Senkov, G B Wilks, D B Miracle, C P Chuang and P K Liaw, Refractory high-entropy alloys, *Intermetallics* **18** 1758 (2010). doi:10.1016/j.intermet.2010.05.014
- [23] E P George, D Raabe and R O Ritchie, High-entropy alloys, *Nat. Rev. Mater.* **4** 515 (2019). doi:10.1038/s41578-019-0121-4
- [24] D B Miracle and O N Senkov, A critical review of high-entropy alloys and related concepts, *Acta Mater.* **122** 448 (2017). doi:10.1016/j.actamat.2016.08.081
- [25] O N Senkov, D B Miracle, K J Chaput and J-P Couzinié, Development and exploration of refractory high-entropy alloys — a review, *J. Mater. Res.* **33** 3092 (2018).

- [26] E E Bloom, The challenge of developing structural materials for fusion power systems, *J. Nucl. Mater.* **258–263** 7 (1998).
- [27] S J Zinkle and G S Was, Materials challenges in nuclear energy, *Acta Mater.* **61** 735 (2013). doi:10.1016/j.actamat.2012.11.004
- [28] S J Zinkle and L L Snead, Designing radiation-resistance in structural materials for fusion energy, *Annu. Rev. Mater. Res.* **44** 241 (2014). doi:10.1146/annurev-matsci-070813-113627
- [29] P K Romano, N E Horelik, B R Herman, A G Nelson, B Forget and K Smith, OpenMC: a state-of-the-art Monte Carlo code for research and development, *Ann. Nucl. Energy* **82** 90 (2015). doi:10.1016/j.anucene.2014.07.048
- [30] J-C Sublet, J W Eastwood, J G Morgan, M R Gilbert, M Fleming and W Arter, FISPACT-II: an advanced simulation system for activation, transmutation and material modelling, *Nucl. Data Sheets* **139** 77 (2017). doi:10.1016/j.nds.2017.01.002
- [31] P I Ford, *Kronos Physics De-Risking Register*, v8, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22133057
- [32] P I Ford, *First-principles near-neutral low-activation refractory alloys for fusion plasma-facing components: formation-energy screening and entropy-stabilized composition-of-matter design*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645744. Official page: kronosfusionenergy.com/publications/low-activation-alloys.html.
- [33] P I Ford, *A tantalum- and rhenium-free low-activation alloy family for first wall, divertor and direct-energy-converter electrodes: a first-principles composition-of-matter study*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645891. Official page: kronosfusionenergy.com/publications/a-tantalum-and-rhenium-free-low-activation-alloy-family.html.
- [34] P I Ford, *A radiation-tolerant refractory high-entropy alloy for fusion plasma-facing components: microstructural stability and a scalable processing route*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645868. Official page: kronosfusionenergy.com/publications/a-radiation-tolerant-refractory-high-entropy-alloy-for-.html.
- [35] P I Ford, *Component activation, waste-class routing, and decommissioning of a low-activation compact spherical-tokamak breeder: no isotope beyond a century's half-life*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645846. Official page: kronosfusionenergy.com/publications/component-activation-waste-class-routing-and-decommissi.html.
- [36] P I Ford, *First-wall and plasma-facing-component lifetime under 14 MeV loading: displacement damage, bubble swelling and erosion in a compact spherical-tokamak breeder*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645774. Official page: kronosfusionenergy.com/publications/first-wall-neutron-damage-lifetime.html.
- [37] P I Ford, *Quantum chemistry beyond DFT for low-activation fusion alloys: ADAPT-VQE correlation energies for composition-of-matter screening with fault-tolerant resource estimates*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645929. Official page: kronosfusionenergy.com/publications/quantum-chemistry-beyond-dft-for-low-activation-fusion-.html.
- [38] P I Ford, *Global sensitivity and uncertainty quantification across the breeder and burner design planes: a 40,000-sample Monte-Carlo ranking of the frozen anchors*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645893. Official page: kronosfusionenergy.com/publications/global-sensitivity-and-uncertainty-quantification-acros.html.